

TA 9.1 A 13.56MHz CMOS RF Identification Transponder Integrated Circuit With A Dedicated CPU

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Radio-frequency identification (RFID) technology is attracting attention for its growing market in telephony and transport fare collection systems, which are replacing magnetic debit card technology, as well as factory automation and access control systems. In the former systems, the data rate is expected to exceed 100kb/s, higher than previous results [1, 2]. Complex functions, such as anti-collision and authentication, are indispensable in transponder ICs in spite of the fact that their addition increases power consumption. To achieve the high data rate along with high magnetic field emission from interrogators, the 13.56MHz ISM band is appropriate for RF power and data transmission. These RF and analog circuits for a 13.56MHz RFID transponder IC are associated clocking and anti-collision techniques controlled by a dedicated CPU.

Figure 9.1.1 is the transponder IC block diagram. Data between interrogator and transponder are transmitted in half-duplex mode. The resonant capacitor C_R and supply capacitor C_L are implemented on-chip, and the off-chip component is an antenna coil L_R . The instruction format of the dedicated CPU is represented by 12b: 4b Op code, 4b destination register address, and 4b source register address. The instruction set has 29 operations including an immediate addressing mode. Registers in the CPU are organized as a program counter, an immediate register, an I/O register and 13 general purpose registers. The demodulated data from the RF block and modulation data from the CPU are transferred through the I/O register. Data transfer between memory (EEPROM and ROM) and registers is operated by LOAD/STORE instructions, in which the memory address field refers to a register.

Table 9.1.1 lists the RF data communication interface between interrogator and transponder. Since the modulation index is 10%, RF power and clock are continuously supplied to transponders. Two data rates, 212kb/s and 106kb/s, can be selected by a CPU program in any sequence of data transmission.

Figure 9.1.2 shows the power control circuit of the transponder IC. The full wave rectifier consists of transistors N1-N2 and P1-P6. The diode-connected pMOS transistors P1-P2 supply RF power to the primary power supply node VDH; while P3-P6 connect n-well potential to a higher terminal of either RF1/2 and VDH. The merit of using pMOS over nMOS transistors is that the voltage drop from the RF terminals to VDH is minimized by the elimination of the body effect. nMOS transistors N1-N2 are not diode-connected but act as switches to minimize voltage drop between RF1/2 and GND terminals. The power supply VDD is stabilized by the series 4V regulator. The primary difference between this series regulator and the Zener diode (shunt regulator) is power consumption [2]. The shunt regulator degrades the communication range of RFID systems through its excess power consumption, especially for the case that several transponders in the near-field of an interrogator absorb the applied magnetic field.

The operation of the series regulator and power-on reset circuit is controlled by a bandgap reference. Through the use of zero-threshold nMOS transistors, the output of the bandgap is valid for $VDH > 1.5V$.

Figure 9.1.3 shows the data communication circuit of the transponder IC. In the demodulation block, the high frequency component of VDH is eliminated by the low pass filter (LPF). Since gain of the subsequent feedback amplifier is -1, $V1-V2$ becomes equal to $-(V3-V2)$. These voltage differences, $V3-V2$ and $V1-V2$, are amplified by the following stages, A2 and A3, to the full swing of supply voltage. For example, when V1 falls at the falling edge of modulation, V4 rises to VDD, and V5 falls to GND; consequently, DATA output becomes low. Measured bit error rate of this demodulation circuit is better than 5×10^{-9} . The current consumption of all amplifiers is limited to 6 μ A by current mirrors. In the modulation circuit, the load switching circuit is separated from rectifiers to enhance the modulation index. One important feature of transponder ICs with a CPU is the controllability of power consumption. When a transponder transmits data or writes data to EEPROM, the power consumed by the CPU and ROM must be lowered to compensate the additional power consumed by the load switching or EEPROM, otherwise, the communication range is degraded. Power consumption of the CPU and ROM is controlled with the clock frequency-control circuit as shown Figure 9.1.4. In the data transmission or EEPROM write sequence, the lower-frequency clock is selected, since fewer CPU executions are required. The clock-select signal from CPU is synchronized by the lower-frequency clock to remove an anomalous short pulse in the resulting clock waveform.

Anti-collision is a method of identifying all transponders in the inquiry area of an interrogator. Figure 9.1.5 illustrates an optimized binary tree search algorithm for the anti-collision ID readout. The interrogator deduces an ID bit from response timing of the transponders, and repeats the sequence for the entire ID code. In the sequence shown in Figure 9.1.5, 2b rather than 1b are searched at once to reduce the entire transaction time. Associated with each 2b, 4 time slots (ID = "00", "01", "10", and "11") are assigned for the transponder response. In Figure 9.1.5, transponders A and C are responding in the first slot regarding their ID of "00". After detecting these responses, the interrogator transmits ACK signal to deactivate all transponders that did not respond. In the subsequent sequence, transponder A responds in the third slot, and the interrogator acknowledges the existence of transponder whose ID is "1000".

A serious issue in this anti-collision sequence is that the transponder response is canceled when the phases of two responses are in opposition. This cancellation results from the asynchronous start of the response, and is resolved by scrambling its phases. In Figure 9.1.5, the response is divided into 4 sub-slots and the phase in the sub-slot is modulated according to a random number generated in each transponder. As shown, the interrogator can detect the response at t_3 , even when other responses are cancelled.

Figure 9.1.6 is a micrograph of the transponder chip. The 11mm² chip is fabricated in a 0.8 μ m CMOS technology. The power dissipation is <2mW for 2.6V supply. The communication range with 1W interrogator and card-size antenna is 10cm. A prototype communication program is written to EEPROM with a program-development tool including an assembler and a programming board.

References:

- [1] Friedman, D. et al., "A Low-Power CMOS Integrated Circuit for Field-Powered Radio Frequency Identification Tags" ISSCC Digest of Technical Papers, pp. 294-295, Feb., 1997.
- [2] Kaiser, U. and Steinhagen W., "A Low-Power Transponder IC for High-Performance Identification Systems" IEEE Journal of Solid-State Circuits, vol. 30, no. 3, pp. 306-310, Mar., 1995. S. Masui, E. Ishii, T. Iwawaki, Y. Sugawara, and K. Sawada, 1999 ISSCC Manuscript.

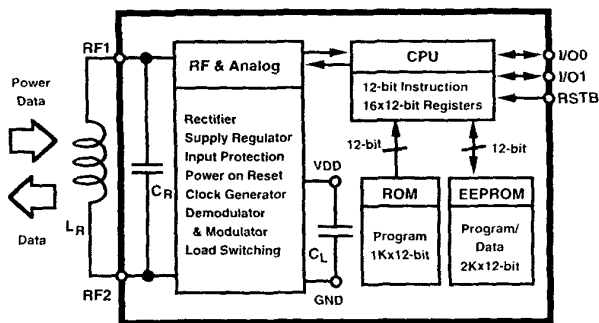


Figure 9.1.1: RFID transponder IC block diagram.

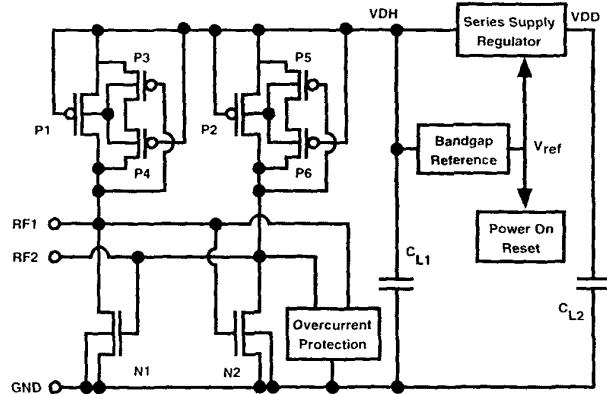


Figure 9.1.2: Power control circuit.

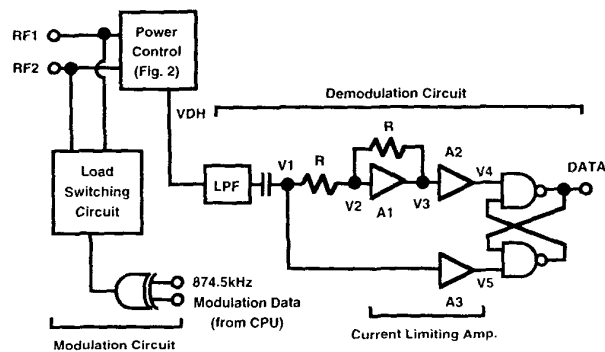


Figure 9.1.3: Data communication circuit.

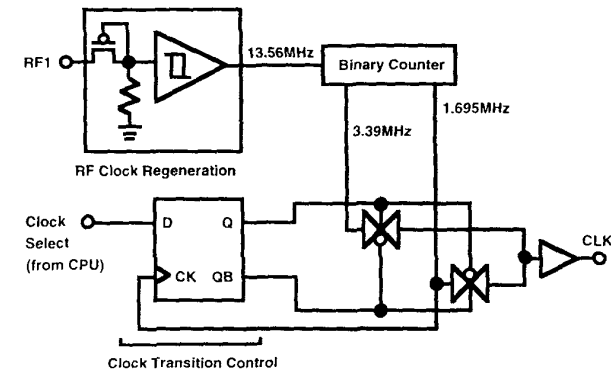


Figure 9.1.4: Clock frequency control circuit.

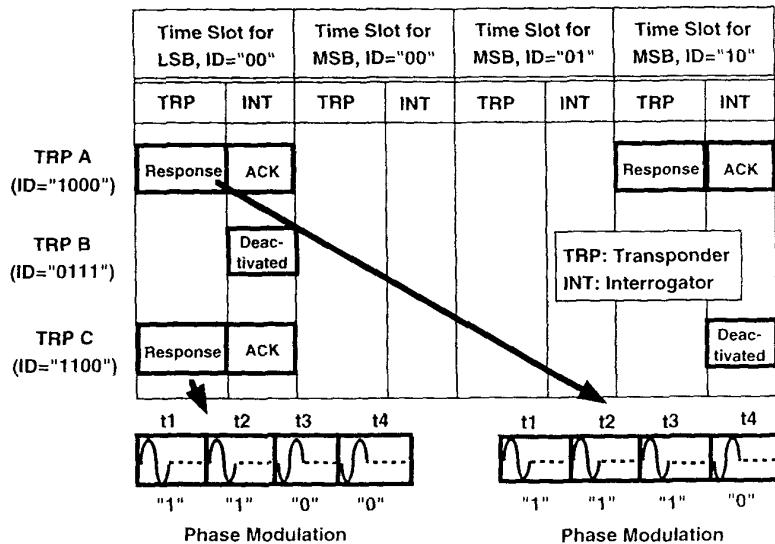


Figure 9.1.5: Binary tree search algorithm for anti-collision ID readout.

Interrogator to Transponder	
Carrier Frequency f_c :	13.56MHz $\pm$ 7kHz
Modulation Type and Index:	ASK 10%
Data Rate:	212kbps ($f_c/64$) / 106kbps ($f_c/128$)
Bit Representation:	NRZ
Transponder to Interrogator	
Applied Magnetic Field Modulation:	Load Switching
Subcarrier Frequency:	847.5kHz
Subcarrier Modulation Type:	BPSK
Data Rate:	212kbps ($f_c/64$) / 106kbps ($f_c/128$)
Bit Representation:	NRZ

Table 9.1.1: RF data communication interface.